

DERIVATIVES

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INTRODUCTION

In this presentation, we will systematically explore derivatives in three core parts:

- The relationships between limit existence, continuity, and differentiability
- All essential derivative rules (power, product, quotient, and chain rule)
- Three real-life applications of derivatives

1. LIMITS & CONTINUITY

Understanding limits, their existence, and continuity of functions

01.

1. Definition of a Limit

The limit of a function $f(x)$ as x approaches a is the value L that $f(x)$ approaches, written as:

$$\lim f(x) = L \text{ (as } x \rightarrow a \text{)}$$

02.

2. Existence of a Limit

A limit exists at $x = a$ if and only if the left-hand limit equals the right-hand limit:

$$\lim f(x) = \lim f(x) = L \text{ (as } x \rightarrow a^- \text{ and } x \rightarrow a^+ \text{)}$$

03.

3. Continuity at a Point

A function $f(x)$ is continuous at $x = a$ if all three conditions are met:

- $f(a)$ is defined
- The limit of $f(x)$ as x approaches a exists
- The limit equals $f(a)$

PRACTICE

Question 1

Evaluate the limit as x approaches 3: $(x^2 - 9) \div (x - 3)$

Solution Step by Step

Step 1: Factor the numerator

$$x^2 - 9 = (x - 3)(x + 3)$$

Step 2: Simplify the fraction

Cancel $(x - 3)$ from numerator and denominator

We get $x + 3$

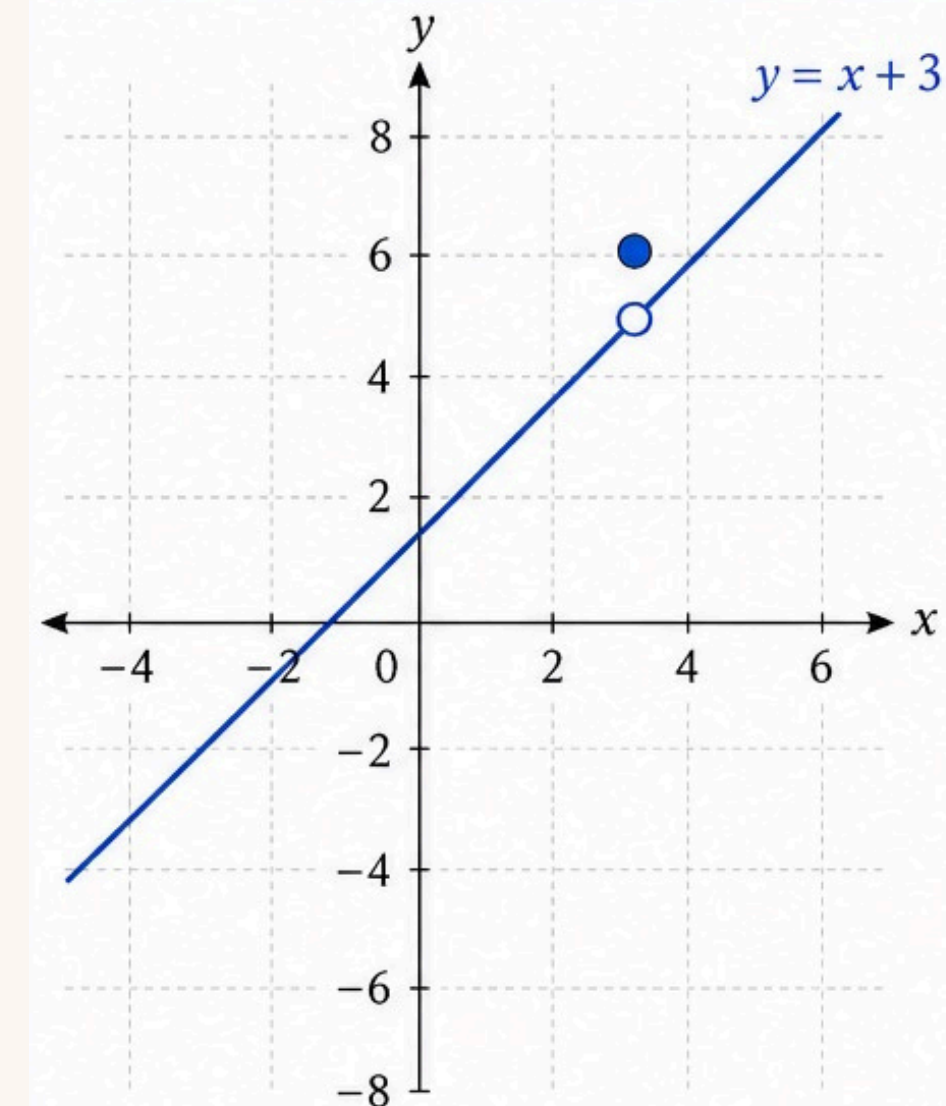
Step 3: Substitute $x = 3$

$$3 + 3 = 6$$

Final Answer: The limit equals 6

Question 1: Limit

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$$



PRACTICE

Question 2

Determine if the function is continuous at $x = 2$

When $x < 2$, $f(x) = x + 1$

When $x = 2$, $f(x) = 4$

When $x > 2$, $f(x) = 2x - 1$

Solution Step by Step

Step 1: Check $f(2)$

$f(2) = 4$, it is defined.

Step 2: Calculate left-hand limit

As x approaches 2 from left: $2 + 1 = 3$

Step 3: Calculate right-hand limit

As x approaches 2 from right: $2(2) - 1 = 3$

Step 4: Compare

Left limit = Right limit = 3, so limit exists.

Step 5: Check continuity

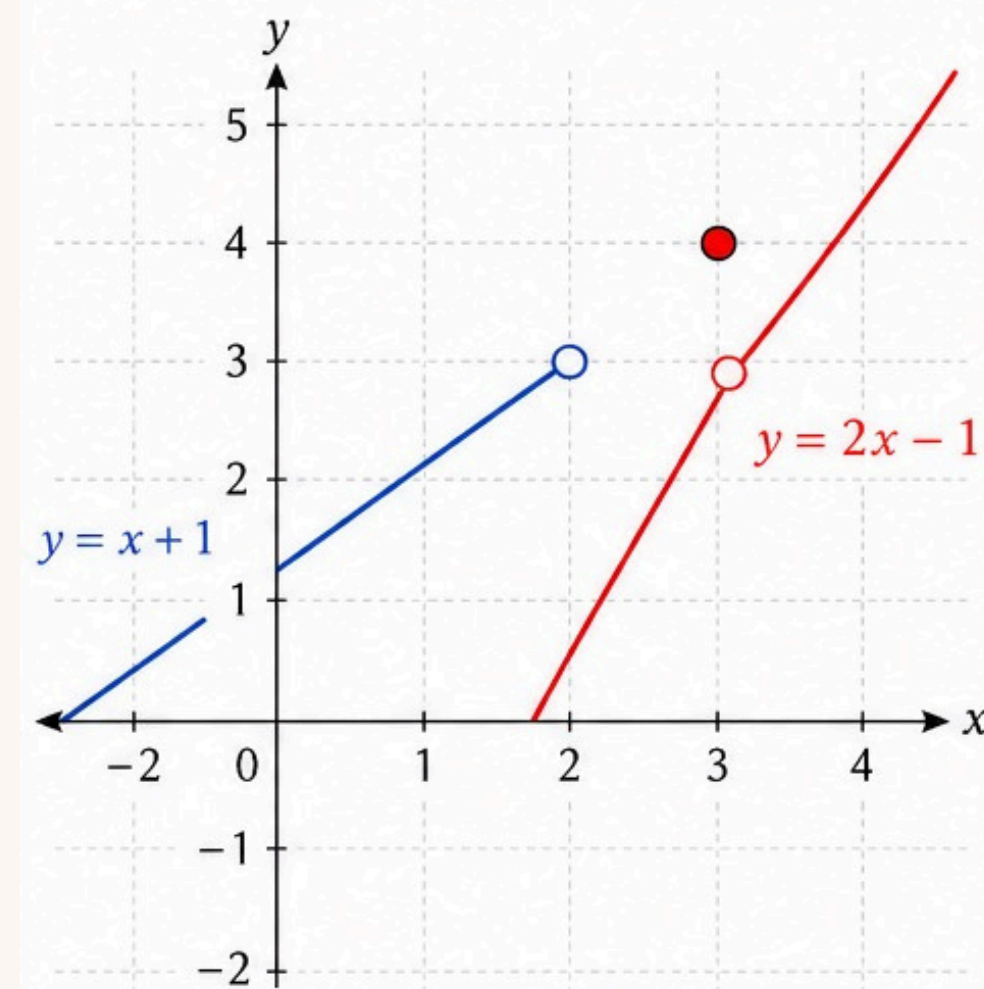
Limit value $3 \neq f(2) = 4$

One condition fails.

Final Answer: The function is NOT continuous at $x = 2$

Question 2: Continuity

$$f(x) = \begin{cases} x + 1, & x < 2 \\ 4, & x = 2 \\ 2x - 1, & x > 2 \end{cases}$$



PRACTICE

Question 3

Find the value of k to make the function continuous at $x = 1$

When $x < 1$, $f(x) = kx^2$

When $x = 1$, $f(x) = 3$

When $x > 1$, $f(x) = x + 2$

Solution Step by Step

Step 1: Find right-hand limit at $x = 1$

$$1 + 2 = 3$$

Step 2: Find left-hand limit at $x = 1$

$$k \times (1)^2 = k$$

Step 3: For continuity

$$\text{Left limit} = \text{Right limit} = f(1)$$

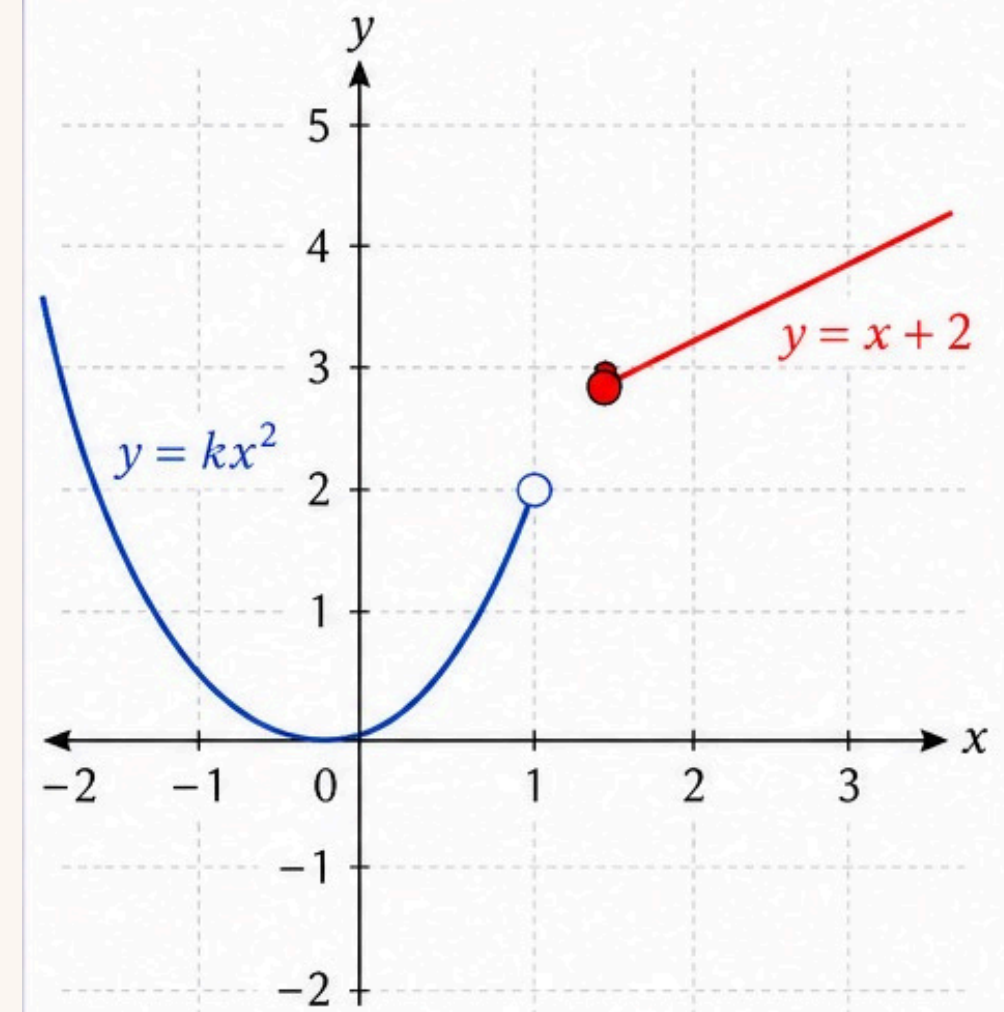
Step 4: Solve for k

$$k = 3$$

Final Answer: $k = 3$

Question 3: Continuity

$$f(x) = \begin{cases} kx^2, & x < 1 \\ 3, & x = 1 \\ x + 2, & x > 1 \end{cases}$$



2. DIFFERENTIABILITY

1. Definition of Differentiability

A function f is differentiable at $x = a$ if the limit exists:

$$f'(a) = \lim [f(a+h) - f(a)] / h \text{ as } h \rightarrow 0$$

2. Geometric Meaning

The derivative is the slope of the tangent line at $x = a$.

If the slope exists, the function is differentiable.

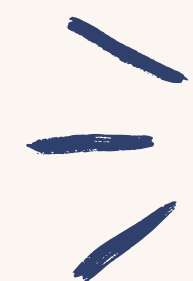
3. Differentiability and Continuity

- If differentiable at a point, then continuous at that point.
- Continuous does NOT guarantee differentiable.

4. Three Non-Differentiable Cases

- Discontinuity
- Corner / sharp point (like $y = |x|$)
- Vertical tangent (infinite slope)





PRACTICE



Question 1

Determine whether the function is differentiable at $x = 1$

$$f(x) = x^2, \text{ for } x \leq 1$$

$$f(x) = 2x - 1, \text{ for } x > 1$$

Step-by-Step Solution

Step 1: Check continuity at $x = 1$

$$\text{Left limit: } 1^2 = 1$$

$$\text{Right limit: } 2(1) - 1 = 1$$

$$f(1) = 1$$

The function is continuous at $x = 1$.

Step 2: Calculate the left-hand derivative (LHD)

$$\text{For } x \leq 1: f(x) = x^2$$

$$\text{Derivative: } f'(x) = 2x$$

$$\text{At } x = 1: \text{LHD} = 2(1) = 2$$

Step 3: Calculate the right-hand derivative (RHD)

$$\text{For } x > 1: f(x) = 2x - 1$$

$$\text{Derivative: } f'(x) = 2$$

$$\text{At } x = 1: \text{RHD} = 2$$

Step 4: Compare LHD and RHD

$$\text{LHD} = \text{RHD} = 2$$

Final Answer: The function is differentiable at $x = 1$.

Question 2

Determine whether the function is differentiable at $x = 0$

$$f(x) = |x|$$

Step-by-Step Solution

Step 1: Check continuity at $x = 0$

$$f(0) = |0| = 0$$

$$\lim_{x \rightarrow 0^-} |x| = 0$$

$$\lim_{x \rightarrow 0^+} |x| = 0$$

The function is continuous at $x = 0$.

Step 2: Calculate the left-hand derivative (LHD)

$$\text{For } x < 0: f(x) = -x$$

$$\text{Derivative: } f'(x) = -1$$

$$\text{At } x = 0: \text{LHD} = -1$$

Step 3: Calculate the right-hand derivative (RHD)

$$\text{For } x > 0: f(x) = x$$

$$\text{Derivative: } f'(x) = 1$$

$$\text{At } x = 0: \text{RHD} = 1$$

Step 4: Compare LHD and RHD

$$\text{LHD} = -1, \text{RHD} = 1$$

They are not equal.

Final Answer: The function is NOT differentiable at $x = 0$.

3.DERIVATIVE RULES

- **Basic Derivative Rules:**
- **Constant Rule**
 - Formula If $f(x)=c$ where c is a constant, then $f'(x)=0$
 - Meaning: The derivative of a constant is 0. For example, the derivative of $y = 5$
- **Power Rule**
 - Formula: If $f(x)=x^n$, then $f'(x)=n \cdot x^{n-1}$
 - Meaning: Bring the exponent down as a multiplier, then subtract 1 from the exponent. This is the most basic tool for differentiation.
- **Sum & Derivative Rules**
 - Formula: If $f(x)=g(x) \pm h(x)$, then $f'(x)=g'(x) \pm h'(x)$
 - Meaning: When differentiating polynomials, differentiate each term separately and then add or subtract the results.



CORE ADVANCED RULES

Rules for Complex & Composite Functions

01

Constant Multiple Rule

02

Product Rule

03

Quotient Rule

04

Chain Rule

CONSTANT MULTIPLE RULE

01

Formula: If $f(x) = c \cdot g(x)$, then $f'(x) = c \cdot g'(x)$

02

Meaning: You can factor out the constant and only differentiate the function that follows it.

PRODUCT RULE

01

Formula: If $f(x) = u(x) \cdot v(x)$, then $f'(x) = u'v + uv'$

02

Meaning: Used when two functions are multiplied. The rule is: "Derivative of the first times the second, plus the first times the derivative of the second."

QUOTIENT RULE

01

Formula: If $f(x) = v(x)/u(x)$, then $f'(x) = [v'(x)u(x) - v(x)u'(x)]/[u(x)]^2$

02

Meaning: Used when two functions are divided. Be careful with the order in the numerator! The rule is: "Derivative of the top times the bottom, minus the top times the derivative of the bottom, all divided by the bottom squared."

CHAIN RULE

01

Formula: If $f(x) = g(h(x))$, then $f'(x) = g'(h(x)) \cdot h'(x)$

02

Meaning: Used for composite functions. Differentiate from the outside to the inside, one layer at a time, then multiply the derivatives together.

THREE EXAMPLE PROBLEMS

Example 1:

Basic Rules Combined (Power + Sum/Difference + Constant Multiple)

Find the derivative of the function

$$f(x) = 3x^4 - 2x^3 + 5x - 7$$

Step-by-Step Solution:

Apply the Sum/Difference Rule: $f'(x) = d/dx(3x^4) - d/dx(2x^3) + d/dx(5x) - d/dx(7)$

Apply the Constant Multiple Rule: $f'(x) = 3 \cdot d/dx(x^4) - 2 \cdot d/dx(x^3) + 5 \cdot d/dx(x) - 7 \cdot d/dx(1)$

Apply the Power Rule and Constant Rule: $d/dx(x^4) = 4x^3$, $d/dx(x^3) = 3x^2$, $d/dx(x) = 1$, $d/dx(1) = 0$

Substitute and simplify: $f'(x) = 3(4x^3)$

$$- 2(3x^2) + 5(1) - 0 \quad f'(x) = 12x^3 - 6x^2 + 5$$

Example 2:

Comprehensive Application (Product + Chain Rule)

Question: Find the derivative of the function

$$f(x) = x^2 * \sin(3x)$$

Apply the Product Rule:

Let $u(x) = x^2$ and $v(x) = \sin(3x)$. $f'(x) = u'(x)v(x) + u(x)v'(x)$

Compute $u'(x)$ using the Power Rule: $u'(x) = 2x$

Compute $v'(x)$ using the Chain Rule: $v'(x) = \cos(3x) \cdot d/dx(3x) = \cos(3x) \cdot 3 = 3\cos(3x)$

Substitute into the formula: $f'(x) = (2x)\sin(3x) + (x^2)(3\cos(3x))$ $f'(x) = 2x\sin(3x) + 3x^2\cos(3x)$

EXAMPLE 3: QUOTIENT RULE + CHAIN RULE

QUESTION: FIND THE DERIVATIVE OF THE FUNCTION

$$F(X) = (2X - 1)^3 / (X + 4)$$

APPLY THE QUOTIENT RULE: LET $U(X) = (2X - 1)^3$ AND $V(X) = X + 4$. $F'(X) = [U'(X)V(X) - U(X)V'(X)] / [V(X)]^2$

COMPUTE $U'(X)$ USING THE CHAIN RULE:

$$U(X) = 3(2X - 1)^2 \cdot D/DX(2X - 1) = 3(2X - 1)^2 \cdot 2 = 6(2X - 1)^2$$

COMPUTE $V'(X)$ USING THE POWER RULE: $V'(X) = 1$

SUBSTITUTE INTO THE FORMULA:

$$F'(X) = [6(2X - 1)^2(X + 4) - (2X - 1)^3(1)] / (X + 4)^2$$



4. THREE APPLICATIONS OF DERIVATIVES TO REAL LIFE



01.

Application 1: Kinematics
(Displacement, Velocity,
and Acceleration)

03.

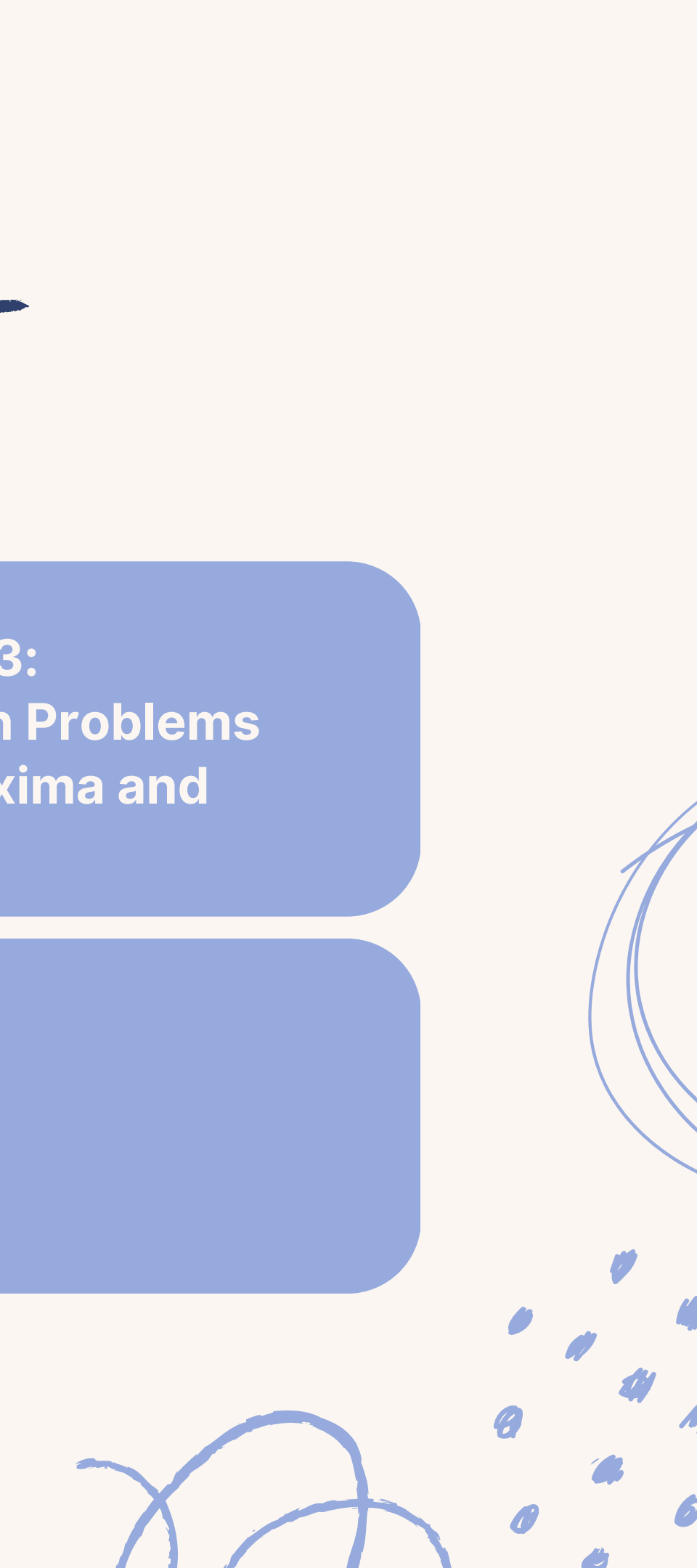
Application 3:
Optimization Problems
(Finding Maxima and
Minima)

02.

Application 2:
Economics (Marginal
Cost / Marginal
Revenue)

04.

Practise



APPLICATION 1: KINEMATICS (DISPLACEMENT, VELOCITY, AND ACCELERATION)

Core Concept: The first derivative of displacement with respect to time is velocity, and the second derivative is acceleration. Derivatives can describe the instantaneous state of motion of an object.

Formulas:

Displacement function: $s(t)$

Velocity: $v(t) = s'(t) = ds/dt$

Acceleration: $a(t) = v'(t) = s''(t) = d^2s/dt^2$

Practical Example:

The displacement function of an object is $s(t) = 3t^2 + 2t$ (Unit: meters, t in seconds).

Velocity: $v(t) = s'(t) = 6t + 2$

Acceleration: $a(t) = v'(t) = 6 \text{ m/s}^2$

This indicates the object is moving with a constant acceleration of 6 m/s^2 . The instantaneous velocity at $t = 2$ seconds is $v(2) = 14 \text{ m/s}$.

APPLICATION 2: ECONOMICS (MARGINAL COST / MARGINAL REVENUE)

Core Concept: The derivative of the cost function with respect to quantity is the marginal cost, representing the change in total cost for producing one additional unit of a product.

Formulas:

Total Cost function: $C(q)$ (where q is quantity)

Marginal Cost: $MC(q)=C'(q)=dC/dq$

Practical Example:

The cost function for a factory producing q items is $C(q)=0.1q^2+5q+200$ (Unit: Yuan).

Marginal Cost: $MC(q)=C'(q)=0.2q+5$

When the quantity is 100 items, the marginal cost $MC(100)=25$ Yuan, meaning the production cost of the 101st item is approximately 25 Yuan.

APPLICATION 3: OPTIMIZATION PROBLEMS (FINDING MAXIMA AND MINIMA)

Core Concept: By finding the critical points of a function through differentiation, we can solve practical optimization problems such as "maximum profit," "minimum cost," and "maximum area."

Solution Steps:

1. Establish the objective function (e.g., profit function $P(x)$).
2. Find the first derivative $P'(x)$.
3. Set $P'(x)=0$ to find critical points.
4. Use the second derivative to determine if it is a local maximum or minimum.

Practical Example:

Use a 10-meter-long fence to enclose a rectangular vegetable garden and find the maximum area.

Let the length be x , the width be $5-x$. The area function is:

$$A(x)=x(5-x)=5x-x^2$$

$$\text{Differentiate: } A'(x)=5-2x$$

$$\text{Set } A'(x)=0, \text{ we get } x=2.5$$

The second derivative $A''(x)=-2<0$, indicating a local maximum at $x=2.5$.

The maximum area is 6.25m^2 .



THANK YOU
VERY MUCH!

